PRODUCTS AND DIRECT SUMS OF SHIFTS AND DISTINGUISHED TRIANGLES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1002-1020, following the statement at 978-1000.

Editorial note. No omissions: (2)(b) is explicitly marked dual to the executed (2)(a).

Allowed standard prerequisites (editorial).

- Special triangles criterion. remark-special-triangles (same chapter, cited at ~1012 and ~1015). A map of special triangles with identity first two entries has invertible third entry.

Source reference. Original remark 09WA: sources/stacks-derived/pinned-original-source.tex, source lines 673–673. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a pre-triangulated category. Let I be a set.*

(1) *Let X_i, $i \in I$ be a family of objects of $\mathcal{D}$.*
   (a) *If $\prod X_i$ exists, then $(\prod X_i)[1] = \prod X_i[1]$.*
   (b) *If $\bigoplus X_i$ exists, then $(\bigoplus X_i)[1] = \bigoplus X_i[1]$.*

(2) *Let $X_i \to Y_i \to Z_i \to X_i[1]$ be a family of distinguished triangles of $\mathcal{D}$.*
   (a) *If $\prod X_i$, $\prod Y_i$, $\prod Z_i$ exist, then $\prod X_i \to \prod Y_i \to \prod Z_i \to \prod X_i[1]$ is a distinguished triangle.*
   (b) *If $\bigoplus X_i$, $\bigoplus Y_i$, $\bigoplus Z_i$ exist, then $\bigoplus X_i \to \bigoplus Y_i \to \bigoplus Z_i \to \bigoplus X_i[1]$ is a distinguished triangle.*

Proof. Part (1) is true because [1] is an autoequivalence of $\mathcal{D}$ and because direct sums and products are defined in terms of the category structure. Let us prove (2)(a). Choose a distinguished triangle $\prod X_i \to \prod Y_i \to Z \to \prod X_i[1]$. For each j we can use TR3 to choose a morphism $p_j : Z \to Z_j$ fitting into a morphism of distinguished triangles with the projection maps $\prod X_i \to X_j$ and $\prod Y_i \to Y_j$. Using the definition of products we obtain a map $\prod p_i : Z \to \prod Z_i$ fitting into a morphism of triangles from the distinguished triangle to the triangle made out of the products. Observe that the "product" triangle $\prod X_i \to \prod Y_i \to \prod Z_i \to \prod X_i[1]$ is special in the terminology of Remark 09WA because products of exact sequences of abelian groups are exact. Hence Remark 09WA shows that the morphism of triangles is an isomorphism and we conclude by TR1. The proof of (2)(b) is dual. □

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.